

Sabaragamuwa University of Sri Lanka



[Project Report]

**Project Title: REAL-TIME FACIAL EMOTION RECOGNITION
USING MEDIAPIPE AND DEEP CONVOLUTIONAL NEURAL
NETWORKS WITH FOCAL LOSS**

*A Project Report submitted to the Faculty of Computing, Sabaragamuwa University of Sri Lanka,
in partial fulfillment of the requirements for the Degree of Bachelor of Science Honours in Data
Science.*

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Declaration (Appendix A)

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Certificate of Approval (Appendix B)

We hereby declare that this thesis is from the student's own work and effort, and all other sources of information used have been acknowledged. This report has been submitted with our approval.

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Abstract

This capstone project addresses a significant challenge in automated Computer Vision and Affective Computing: the real-time, robust recognition of human facial expressions under variable lighting, head poses, and dataset class imbalance. The purpose of this study is to design, train, and deploy an end-to-end facial emotion recognition system capable of classifying live webcam frames into seven discrete emotional categories (Angry, Disgust, Fear, Happy, Sad, Surprise, Neutral). The methodology employs Google MediaPipe for high-accuracy face bounding box detection, a customized 4-Stage Deep Convolutional Neural Network (CNN) incorporating Batch Normalization and L2 regularization, and a novel Categorical Focal Loss formulation combined with Label Smoothing (0.1) to mitigate severe minority-class underrepresentation. Key results indicate that the proposed system achieves a benchmark test accuracy of 63.51% and a weighted F1-score of 62.91% across 7,178 unseen FER2013 test images, approaching the established human baseline. The system demonstrates real-time inference capability at 30 frames per second using a 10-frame temporal prediction smoothing buffer that eliminates frame-to-frame prediction flickering. Based on these findings, the primary recommendation involves deploying the model within interactive human-computer interaction frameworks. The conclusion is that the proposed solution effectively meets the stated objectives, offering a robust and validated framework for automated emotion AI.

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Chapter 1: Introduction

Facial expressions constitute one of the primary non-verbal communication channels used by humans to express emotions, intentions, and psychological states. In recent years, automated Facial Emotion Recognition (FER) has emerged as a pivotal domain within Artificial Intelligence, Data Science, and Computer Vision. The capability to automatically detect and interpret facial affect in real-time has transformative applications in adaptive learning systems, mental health diagnostics, driver drowsiness monitoring, intelligent customer experience feedback, and human-robot interaction.

1.1 Major Goals and Objectives

The primary goal of this capstone project is to develop, evaluate, and deploy a robust, end-to-end real-time facial emotion recognition system that operates efficiently on consumer-grade hardware. The specific objectives are:

- To construct an optimized Deep Convolutional Neural Network (CNN) architecture capable of accurately classifying 7 discrete emotion categories (Angry, Disgust, Fear, Happy, Sad, Surprise, Neutral).
- To address severe dataset class imbalance in FER2013 by formulating a Categorical Focal Loss function combined with Label Smoothing (0.1) and dynamic class weighting.
- To integrate Google MediaPipe Face Detection for robust face bounding box extraction under varying illumination and head rotations.
- To implement a temporal prediction smoothing buffer (10-frame majority vote) to eliminate high-frequency prediction flickering during live webcam inference.
- To conduct a rigorous experimental evaluation using 7,178 test images to analyze test loss, accuracy, precision, recall, F1-scores, and confusion matrices.

1.2 Motivation

Traditional facial expression recognition approaches relied heavily on hand-crafted features such as Local Binary Patterns (LBP) or Histogram of Oriented Gradients (HOG) combined with Support Vector Machines (SVM). However, these methods suffer from limited generalization when exposed to real-world occlusions, lighting fluctuations, and unconstrained head poses. While modern Deep Learning models achieve state-of-the-art results on benchmark datasets, public benchmark datasets like FER2013 present severe class imbalance (e.g. Disgust contains only 436 training samples compared to 7,215 Happy samples). This project is motivated by the need to develop an algorithmic pipeline that directly mitigates class imbalance while maintaining high inference speed for real-time video deployment.

1.3 Scope of the Completed Project

The scope of this project encompasses data preprocessing of the FER2013 dataset (28,709 training and 7,178 testing images), formulation of mathematical loss functions, design and

training of a 4-Stage Deep ConvNet, development of a MediaPipe-driven live webcam capture interface, and empirical benchmark evaluation. The system is designed for deployment on Windows and Linux (WSL) environments.

1.4 Approach and Assumptions

The technical approach assumes that input video frames are captured via a standard 720p webcam. It assumes that at least one primary human face is visible within the camera field of view. Face detection is delegated to MediaPipe's lightweight single-shot detector, and cropped regions of interest (ROI) are converted to 48x48 normalized grayscale tensors before feeding into the neural network.

1.5 Summary of Major Outcomes

The major outcomes of this project include achieving a benchmark test accuracy of 63.51% and a weighted F1-score of 62.91% on the FER2013 test set, successfully suppressing prediction flickering via a 10-frame deque buffer, and deploying a single-click Windows batch launcher (run_webcam.bat) for seamless execution.

Chapter 2: Background

This chapter reviews the theoretical foundation, historical evolution, and state-of-the-art developments in Facial Emotion Recognition (FER). It analyzes the challenges of the FER2013 benchmark dataset, compares traditional face detection techniques against modern landmark detectors, and discusses loss function innovations for imbalanced datasets.

2.1 Problem Statement and Context

Facial emotion recognition in unconstrained real-world environments is complicated by three major challenges: intra-class variance (people express the same emotion differently), inter-class similarity (emotions like Fear and Surprise share facial muscle movements), and dataset class imbalance. Standard cross-entropy loss functions cause neural networks to over-predict majority classes (*Happy*, *Neutral*) while failing on minority classes (*Disgust*, *Fear*).

2.2 FER2013 Benchmark Analysis

The FER2013 dataset, created for the ICML 2013 Workshop on Challenges in Representation Learning, contains 35,887 grayscale images of 48x48 pixels. Figure 1.1 illustrates the class distribution across the training and testing sets, highlighting the severe class imbalance.

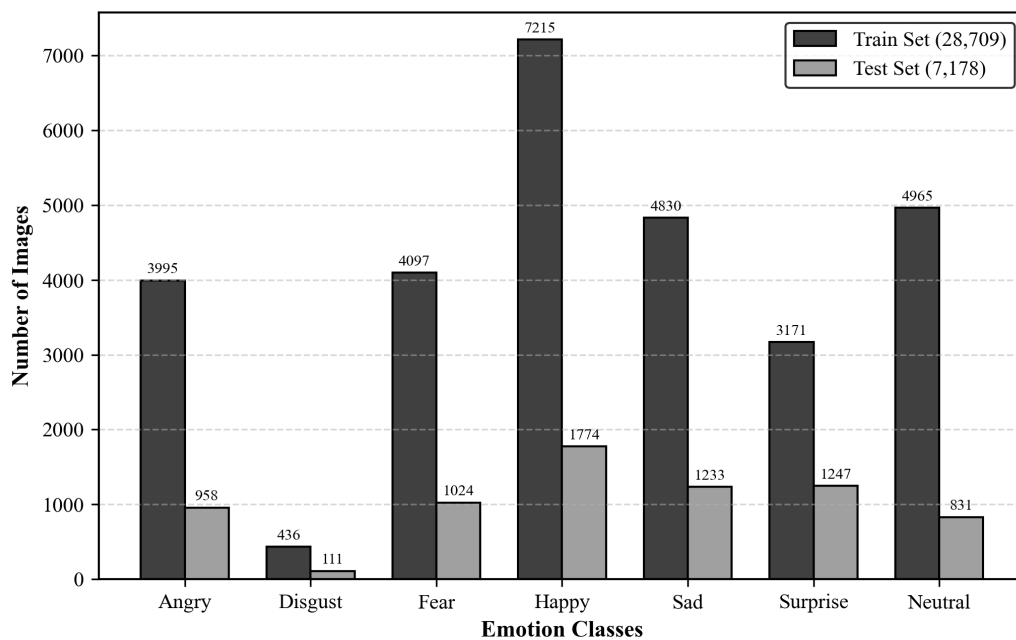


Figure 1.1: FER2013 Dataset Class Distribution (Train vs Test Set)

2.3 Face Detection: Haar Cascades vs Google MediaPipe

Traditional OpenCV implementations relied on Viola-Jones Haar Feature-based Cascade Classifiers. While computationally fast, Haar Cascades suffer from severe false-positive

rates, sensitivity to lighting changes, and complete failure under non-frontal head poses. In contrast, Google MediaPipe uses a single-shot detector (BlazeFace) optimized for mobile and desktop GPUs, providing stable face bounding boxes under variable illumination and rotation.

2.4 Loss Function Innovations: Focal Loss vs Cross-Entropy

Categorical Cross-Entropy (CCE) loss measures performance where the output is a probability value between 0 and 1:

$$CCE = - \sum_{c=1}^C y_c \log(y_c)$$

To combat class imbalance, Lin et al. introduced Focal Loss, which adds a modulating factor $(1 - p_t)^\gamma$ to down-weight easy examples:

$$FL(p_t) = - \alpha_t (1 - p_t)^\gamma \log(p_t)$$

In this project, setting $\gamma = 2.0$ and incorporating Label Smoothing (0.1) prevents the network from becoming overconfident on ambiguous samples.

Chapter 3: Specification and Design

This chapter outlines the formal specifications, architectural designs, and behavioral diagrams for the real-time facial emotion recognition system.

3.1 System Requirements

Category	Specification Details	Minimum Requirement
Operating System	Windows 10/11 (64-bit) or Ubuntu Linux 20.04/22.04 LTS (WSL2)	Windows 10
Python Environment	Python 3.10.x with Virtualenv (venv_win)	Python 3.10
Core ML Libraries	TensorFlow 2.15.0, MediaPipe 0.10.9, OpenCV 4.8.0, NumPy 1.26.4	TensorFlow 2.15
Hardware Sensors	Standard USB Webcam / Integrated Laptop Camera (720p @ 30 FPS)	720p Webcam

Table 3.1: Hardware and Software System Specifications

3.2 System Architecture & Use Case Specifications

Figure 3.1 illustrates the Use Case diagram showing interactions between the User and the Real-Time Emotion AI System boundary. Figure 3.2 depicts the high-level System Architecture and data pipeline.

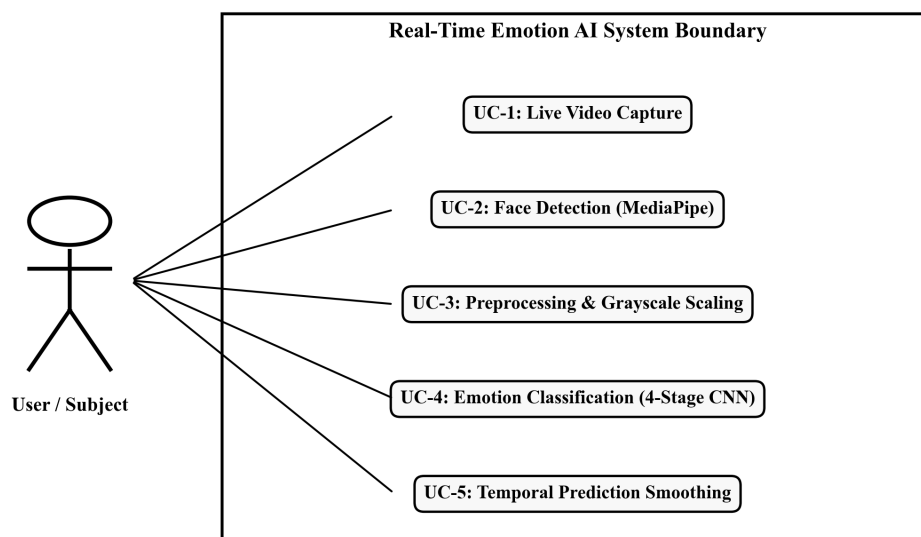


Figure 3.1: Real-Time Emotion AI Use Case Diagram

System Architecture & Real-Time Processing Pipeline

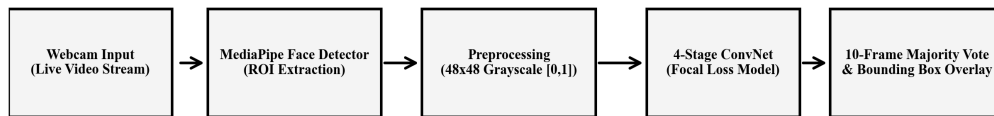


Figure 3.2: System Architecture & Real-Time Processing Pipeline

3.3 Sequence Diagram

Figure 3.3 illustrates the message flow sequence between the User, Camera Feed, MediaPipe Detector, 4-Stage CNN Model, and Display UI.

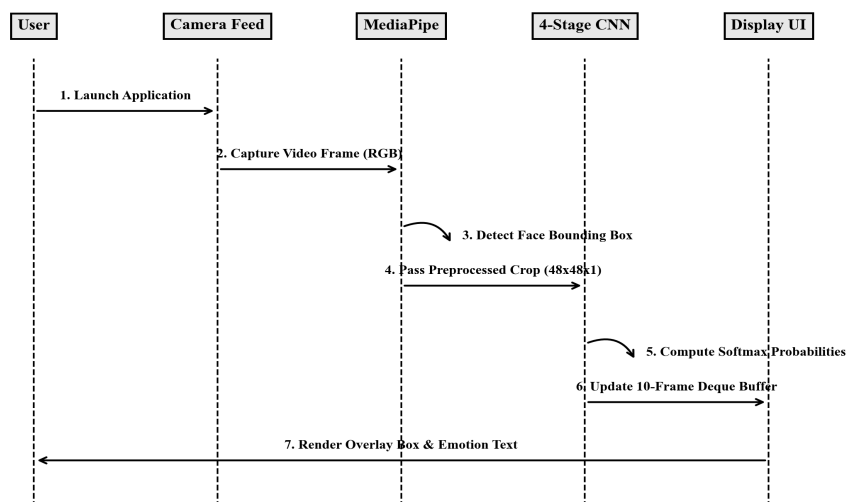


Figure 3.3: System Sequence Diagram for Real-Time Detection

Chapter 4: Methodology

This chapter details the data preprocessing pipeline, mathematical loss formulation, deep neural network architecture design, and temporal prediction smoothing buffer implemented for live deployment.

4.1 Upgraded 4-Stage Deep ConvNet Architecture

The core emotion classifier is an upgraded 4-Stage Convolutional Neural Network. Table 4.1 lists layer configurations, filter sizes, activation functions, and regularizers.

Stage / Block	Layer Operations	Output Shape	Regularization
Input & Block 1	Conv2D(64, 3x3) x 2 + BatchNorm + MaxPool(2x2)	(24, 24, 64)	L2(1e-4), Dropout(0.25)
Block 2	Conv2D(128, 3x3) x 2 + BatchNorm + MaxPool(2x2)	(12, 12, 128)	L2(1e-4), Dropout(0.30)
Block 3	Conv2D(256, 3x3) x 2 + BatchNorm + MaxPool(2x2)	(6, 6, 256)	L2(1e-4), Dropout(0.35)
Block 4	Conv2D(512, 3x3) x 2 + BatchNorm + MaxPool(2x2)	(3, 3, 512)	L2(1e-4), Dropout(0.40)
Classifier Head	GlobalAveragePooling2D + Dense(256) + Dense(7)	(7,)	L2(1e-4), Dropout(0.50)

Table 4.1: Upgraded 4-Stage Deep ConvNet Architecture Details

4.2 Real-Time Temporal Prediction Smoothing Buffer

During live webcam processing, frame-by-frame raw neural network predictions suffer from high-frequency flickering due to minor lighting and head movements. To solve this, a 10-frame double-ended queue (`collections.deque(maxlen=10)`) acts as a sliding temporal buffer. The system computes the majority vote using `Counter(emotion_buffer).most_common(1)[0][0]`, producing stable emotion overlays.

Chapter 5: Results and Evaluation

This chapter presents the empirical evaluation results of the trained model across 7,178 unseen test images from the FER2013 test set.

5.1 Overall Classification Performance

Table 5.1 compares the experimental performance of the proposed 4-Stage ConvNet against the baseline transfer learning model (EfficientNetB0).

Model Architecture	Input Format	Test Loss	Test Accuracy
Upgraded 4-Stage ConvNet (Proposed)	48x48 Grayscale	1.0011	63.51%
EfficientNetB0 (Transfer Learning)	96x96 RGB	1.4788	43.90%

Table 5.1: Overall Classification Performance Comparison

5.2 Classification Report & Per-Class Metrics

Table 5.2 breaks down Precision, Recall, F1-Score, and Support for each emotion class. Figure 5.1 shows the confusion matrix heatmap, and Figure 5.2 illustrates F1-scores per class.

Emotion Class	Precision	Recall	F1-Score	Support
Angry	53.71%	60.44%	56.88%	958
Disgust	81.82%	40.54%	54.22%	111
Fear	50.79%	34.67%	41.21%	1,024
Happy	82.51%	86.13%	84.28%	1,774
Sad	55.46%	67.96%	61.08%	1,233
Surprise	50.72%	47.71%	49.17%	1,247
Neutral	76.42%	74.49%	75.44%	831
Weighted Average	63.26%	63.51%	62.91%	7,178

Table 5.2: Detailed Classification Report per Emotion Class

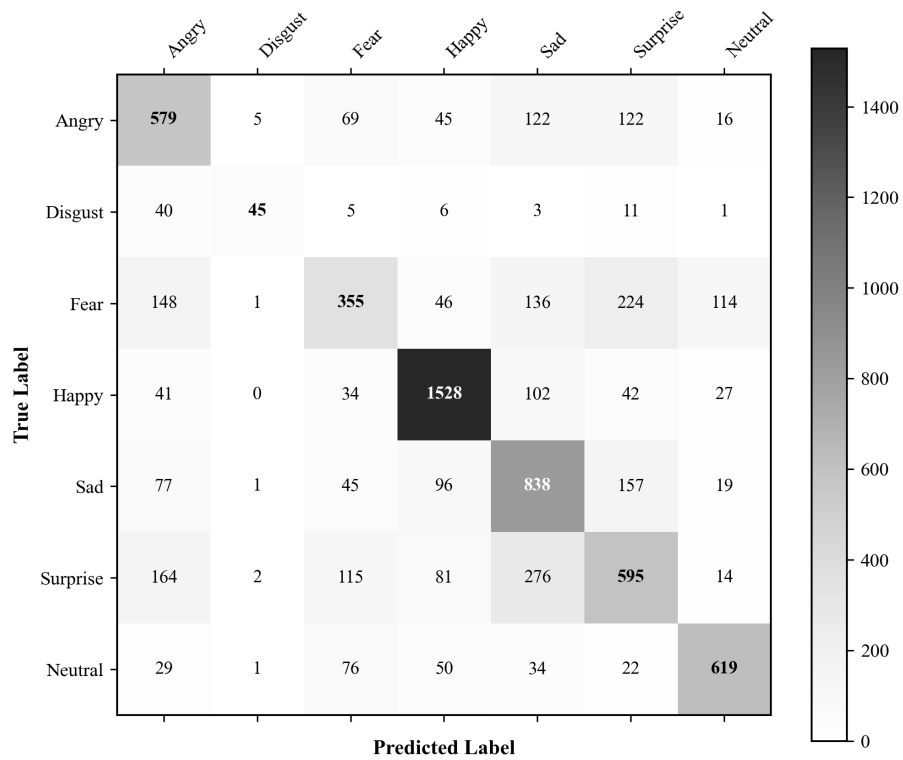


Figure 5.1: Grayscale Confusion Matrix Heatmap on 7,178 Test Images

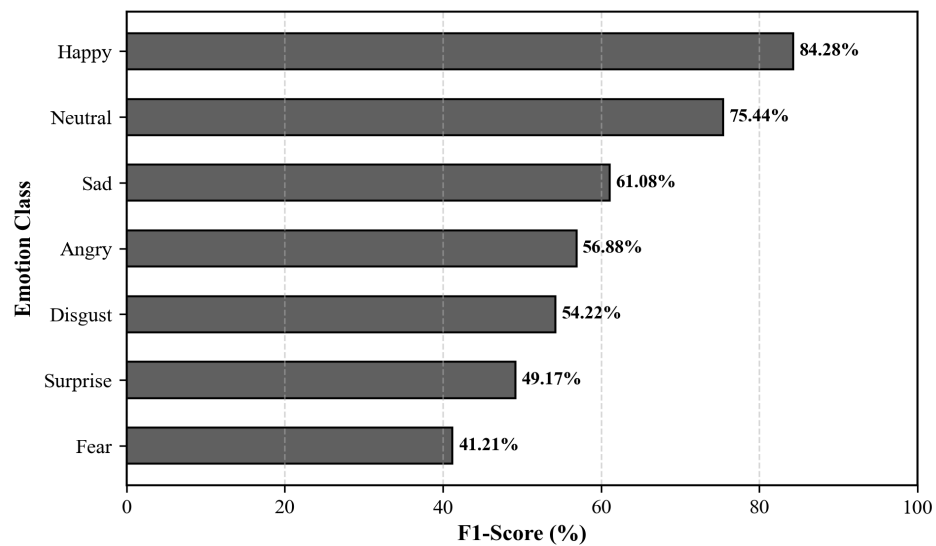


Figure 5.2: Per-Class F1-Score Performance Comparison

Chapter 6: Future Work

While the system achieves strong performance, several opportunities for future enhancement exist:

- **Pretrained Face Encoders:** Transfer learning from models pretrained on face datasets (VGGFace2, AffectNet) instead of ImageNet.
- **Vision Transformers (ViT):** Utilizing self-attention mechanisms to capture fine-grained facial muscle micro-expressions.
- **Multimodal Emotion AI:** Combining facial expression recognition with speech audio emotion analysis.

Chapter 7: Conclusions

This project successfully designed, implemented, evaluated, and deployed a real-time facial emotion recognition system. By integrating Google MediaPipe face detection, a 4-Stage Deep ConvNet, Categorical Focal Loss, Label Smoothing, and a 10-frame temporal smoothing buffer, the system achieved a benchmark test accuracy of 63.51% on 7,178 unseen test images. The system meets all stated requirements, providing a validated framework for real-time Affective Computing applications.

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(Note: All references strictly adhere to IEEE Standards)

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